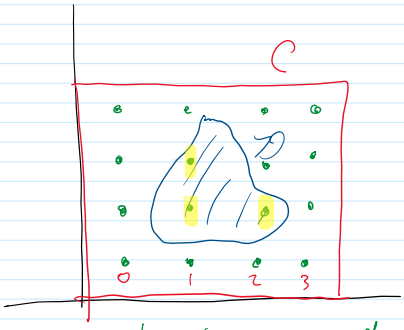
24. Volume Estimation

March 13, 2026 9:10 AM

Problem: region $\mathcal{D} \subset C = [0,1]^{\times d}$

bounded region in $\mathbb{R}^d$ — unit hyper cube



What is the volume of $\mathcal{D}$?

Deterministic Approach (Grid Method)

$K = 4$, $d = 2$, $K^d = 16$, $S = 3$

- Choose K, # of Grid points per row
  - $\Rightarrow$ Total # of points $K^d = n$
  - $\Rightarrow$ # of points in $\mathcal{D}$ is S
- Grid points are $x = (x_1, \dots, x_d)$ where each $x_i \in \left\{ \frac{0 + 1/2}{K}, \frac{1 + 1/2}{K}, \dots, \frac{(K-1) + 1/2}{K} \right\}$
- Error in estimation is

$$\underbrace{\left| \mathrm{Vol}(\mathcal{D}) - \frac{S}{K^d} \right|}_{\text{true} \quad \text{est}} \le \frac{\mathrm{Const}}{K}$$

Surface($\mathcal{D}$)

$\leftarrow O(K^{-1})$ error rate

$\Rightarrow$ increase K and decrease the error

But $n = K^d$ $\therefore$ $K = n^{1/d}$

and our error decreases like $O(n^{-1/d})$

in High dimensions we need alot of points!

Curse of Dimensionality: if we fix an error tolerance $\varepsilon > 0$

then need $n \gtrsim (1/\varepsilon)^d$

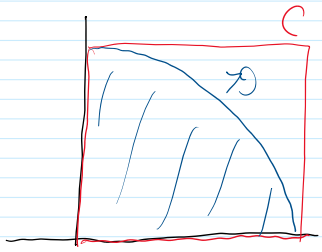
Notation: $\gtrsim$ greater than up to a constant.

Volume of a sphere (Example)

$$\mathcal{D} = \{ x \in \mathbb{R}^d : x_i \ge 0, \ \sum x_i^2 \le 1 \}$$

$$\mathrm{Vol}(\mathcal{D}) = \frac{(\pi/4)^{d/2}}{\Gamma(d/2 + 1)} \longrightarrow 0 \quad \text{as } d \to \infty$$

$$\mathrm{Surf}(\mathcal{D}) = \frac{2(\pi/4)^{d/2}}{\Gamma(d/2)}$$



Consider the relative or normalized error

$$\frac{\left| \mathrm{Vol}(\mathcal{D}) - S/K^d \right|}{} \le \frac{1}{} \frac{\mathrm{Surf}(\mathcal{D})}{} = \varepsilon$$

Question: what is the sample size

$$\frac{|\mathrm{Vol}(\mathcal{D}) - S/K^d|}{\mathrm{Vol}(\mathcal{D})} \le \underbrace{\frac{1}{K}\frac{\mathrm{Surf}(\mathcal{D})}{\mathrm{Vol}(\mathcal{D})}}_{} = \varepsilon$$

What is the sample size to get a relative error $\le \varepsilon$

Do the Math!

$$\varepsilon = n^{-1/d}\frac{\mathrm{Surf}(\mathcal{D})}{\mathrm{Vol}(\mathcal{D})} \Rightarrow n = \left(\frac{d}{\varepsilon}\right)^d$$

Conclusion, for a fixed ε, n grows extremely fast as d increases

if $\varepsilon = 0.001$,

$d =$	2	3	4	5
$\log_{10}(n) =$	6.6	10.4	14.4	18.5

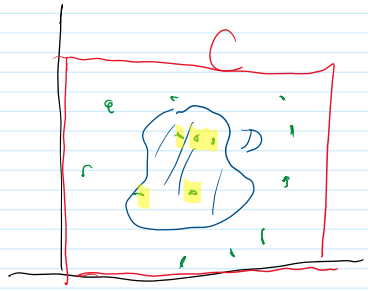
Take away: Can't feasibly solve for $\mathrm{Vol}(\mathcal{D})$ this way!

New idea: Pick points at random instead of a deterministic grid.

- Sample n points in C

  $X_1, \dots, X_n$ where each coordinate $X_{ij} \overset{iid}{\sim} \mathrm{Unif}[0,1]$

  $i = 1, \dots, n$, $j = 1, \dots, d$



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Then, $\phi(x) = \begin{cases} 1, & x \in \mathcal{D} \\ 0, & x \notin \mathcal{D} \end{cases}$

$$\mathrm{Vol}(\mathcal{D}) = \int_{\mathbb{R}^d} \phi(x)\,dx = \int_{\mathbb{R}^d} \frac{1}{n}\sum_{i=1}^n \phi(x_i)\,dx_i \approx \frac{1}{n}\sum_{i=1}^n \phi(x_i)$$

$$= \frac{S}{n} = \frac{(\#\text{ pts in } \mathcal{D})}{(\text{total pts.})}$$

$$\mathbb{E}[\phi(X)] = \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^n \phi(X_i)\right]$$

Key Point: This approach returns a "random" estimate instead of a deterministic estimate.

- Q: what is the mean? (unbiased?)
  Q: what is the variance?
  ↳ How quickly does it go to zero?
  ↳ Confidence interval for the Vol estimate.